Case Study 10: Retail Billing with Multiple Discount Rules

1. Problem Statement

Develop a Python program to generate a retail bill for multiple products. The program should calculate the total price, apply discounts based on the bill amount, calculate taxes according to product categories, and display the final bill.

2. Objectives

- To calculate the cost of multiple products.
- To apply discounts based on the total bill.
- To calculate category-based taxes.
- To display the final payable amount.

3. Assumptions

- The user enters the product name, category, price, and quantity.
- Category essential has 5% tax.
- Category other has 12% tax.
- Bills of ₹1,000 or more receive a 5% discount.
- Bills of ₹2,000 or more receive a 10% discount.
- The maximum discount is ₹300.
- All prices and quantities entered are positive.

Note: These are assumed rules for this assignment, not actual store policies.

4. Algorithm

1. Start.
2. Enter the number of products.
3. For each product, enter its name, category, price, and quantity.
4. Calculate the product total using price × quantity.

5. Calculate tax based on the product category.
6. Calculate the subtotal and total tax.
7. Apply the discount based on the subtotal.
8. Limit the discount to ₹300.
9. Calculate the final bill.
10. Display the subtotal, discount, tax, and final amount.
11. Stop.

5. Flowchart



6. Pseudocode

```
START
INPUT number of products
SET subtotal = 0
SET total_tax = 0

FOR each product
  INPUT name, category, price, quantity
  CALCULATE product_total = price * quantity
  ADD product_total to subtotal

  IF category is essential
    CALCULATE tax = 5% of product_total
  ELSE
    CALCULATE tax = 12% of product_total
  END IF

  ADD tax to total_tax
END FOR

IF subtotal >= 2000
  discount = 10% of subtotal
ELSE IF subtotal >= 1000
  discount = 5% of subtotal
ELSE
```

```
   discount = 0
END IF

IF discount > 300
   discount = 300
END IF

CALCULATE final_bill = subtotal + total_tax - discount
DISPLAY subtotal, total_tax, discount, final_bill
STOP
```

7.Source code

Retail Billing with Multiple Discount Rules.py

```python
n = int(input("Enter number of products: "))
subtotal = 0
total_tax = 0

for i in range(1, n + 1):
    print("\nProduct", i)

    name = input("Enter product name: ")
    category = input("Enter category (essential/other): ")
    price = float(input("Enter price: "))
    quantity = int(input("Enter quantity: "))

    product_total = price * quantity
    subtotal = subtotal + product_total

    if category == "essential":
        tax = product_total * 5 / 100
    else:
        tax = product_total * 12 / 100

    total_tax = total_tax + tax

    print("Product total:", product_total)

if subtotal >= 2000:
    discount = subtotal * 10 / 100
elif subtotal >= 1000:
    discount = subtotal * 5 / 100
else:
    discount = 0

if discount > 300:
```

```
    discount = 0

if discount > 300:
    discount = 300

final_bill = subtotal + total_tax - discount

print("\n----- FINAL BILL -----")
print("Subtotal: Rs.", round(subtotal, 2))
print("Total tax: Rs.", round(total_tax, 2))
print("Discount: Rs.", round(discount, 2))
print("Final amount: Rs.", round(final_bill, 2))
```

8.Test cases

Subtotal	Tax	Discount	Final amount
₹500	₹25	₹0	₹525
₹1000	₹50	₹50	₹1000
₹2500	₹125	₹250	₹2375

9. Output

```
Enter number of products: 1

Product 1
Enter product name: rice
Enter category (essential/other): essential
Enter price: 500
Enter quantity: 2
Product total: 1000.0

----- FINAL BILL -----
Subtotal: Rs. 1000.0
Total tax: Rs. 50.0
Discount: Rs. 50.0
Final amount: Rs. 1000.0

Process finished with exit code 0
```

```
Product 1
Enter product name: rice
Enter category (essential/other): essential
Enter price: 400
Enter quantity: 3
Product total: 1200.0

Product 2
Enter product name: water cans
Enter category (essential/other): essential
Enter price: 20
Enter quantity: 5
Product total: 100.0

----- FINAL BILL -----
Subtotal: Rs. 1300.0
Total tax: Rs. 65.0
Discount: Rs. 65.0
Final amount: Rs. 1300.0
```

10. Conclusion

The program generates a retail bill for multiple products. It calculates product totals, applies category-based taxes and discounts, and displays the final payable amount.